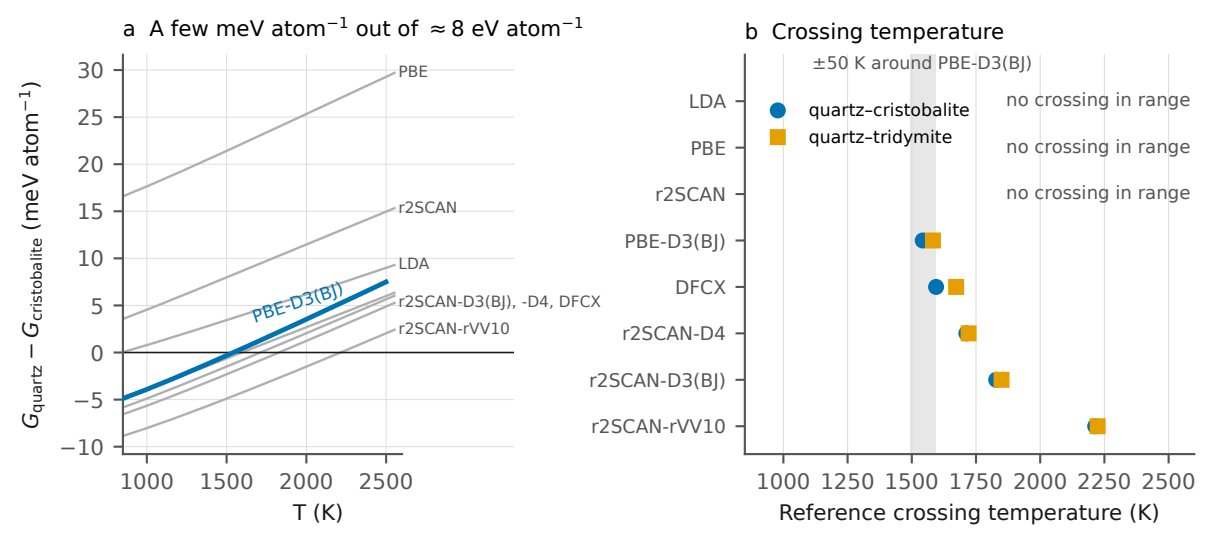
a A few meV atom^{-1} out of ≈ 8 eV atom^{-1}
$G_{quartz} - G_{cristobalite}$ (meV atom^{-1})
30
25
20
15
10
5
0
−5
−10
1000
1500
2000
2500
T (K)
PBE
r2SCAN
LDA
PBE-D3(BJ)
r2SCAN-D3(BJ), -D4, DFCX
r2SCAN-rVV10
b Crossing temperature
±50 K around PBE-D3(BJ)
quartz–cristobalite
quartz–tridymite
LDA
PBE
r2SCAN
PBE-D3(BJ)
DFCX
r2SCAN-D4
r2SCAN-D3(BJ)
r2SCAN-rVV10
no crossing in range
no crossing in range
no crossing in range
1000
1250
1500
1750
2000
2250
2500
Reference crossing temperature (K)